



C23-EE-105

23060

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DEE – FIRST YEAR EXAMINATION

ELECTRICAL ENGINEERING MATERIAL SCIENCE

Time : 3 hours]

[Total Marks : 80

PART—A

3×10=30

- Instructions :**
- (1) Answer **all** questions.
 - (2) Each question carries **three** marks.
 - (3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

- 1. State any six requirements of high resistivity materials.
- 2. Define the terms (a) Hardening and (b) Annealing.
- 3. Distinguish between conductors, insulators and semiconductors.
- 4. List any three of di-electric materials. applications
- 5. Define eddy current loss?
- 6. State Faraday's laws of electro-magnetic induction.
- 7. Draw the field patterns due to (a) Straight current carrying conductor and (b) Solenoid.
- 8. State Len's law.
- 9. State Coulomb's laws of electrostatics.
- 10. Define the terms (a) Electric flux, (b) Flux density related to electrostatics.

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[Contd...

PART—B

10×5=50

- Instructions :** (1) Answer *any five* questions.
(2) Each question carries **ten** marks.
(3) Answers should be comprehensive and the criteria for valuation is the content but not the length of the answer.

- 11.** State the properties and applications of copper and aluminium. 10
- 12.** Distinguish between intrinsic and extrinsic semi-conductors. 10
- 13.** State the effects of the following on P.V.C. (a) Filler, (b) Stabilizer, (c) Plasticizer and (d) Additives. 10
- 14.** State the properties and applications of (a) Impregnated paper and (b) Glass. 10
- 15.** Explain Hysteresis loop with a neat sketch. 10
- 16.** Explain dynamically and statically induced E.M.F. 10
- 17.** (a) Develop an expression for the energy stored in a magnetic field. 5
(b) Compare magnetic circuit with electric circuit in any five aspects. 5
- 18.** (a) Derive an expression for equivalent capacitance when two capacitors are connected in parallel. 5
(b) State different types of capacitors with its uses. 5

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